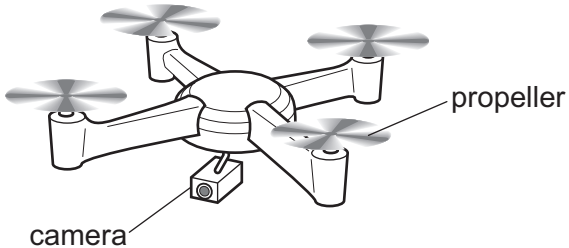
7 A camera drone of mass 1.20 kg hovers at a fixed point above the ground. The drone has four propellers.



In a time of 1.00 s, each propeller pushes a mass of 0.400 kg of air vertically downwards.

Assume that the air above the propellers is stationary.

What is the speed of the air leaving each propeller?

A $0.750\,m\,s^{-1}$ **B** $3.00\,m\,s^{-1}$ **C** $7.36\,m\,s^{-1}$ **D** $29.4\,m\,s^{-1}$